3 **(a)** **(i)** State one difference between a stationary wave and a progressive wave.

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..... [1]

(ii) State why electromagnetic waves can be polarised but sound waves cannot be polarised.

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..... [1]

(b) In Fig. 3.1, T_1 and T_2 are two adjacent transmitters 1.0 m apart with a receiver aerial R at mid-point between them. The transmitters are set up to emit polarised coherent microwaves of wavelength 3.0 cm.

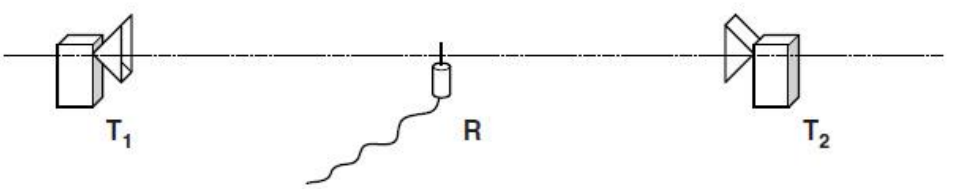

Fig. 3.1

(i) A student observes that the signal at the receiver R falls from maximum to zero when receiver R moved 0.75 cm towards a transmitter.

Explain these observations.

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..... [4]

(ii) With R at the mid-point between T_1 and T_2, the student rotates T_2 through 90° about an axis through T_1 and T_2 as shown in Fig. 3.2.

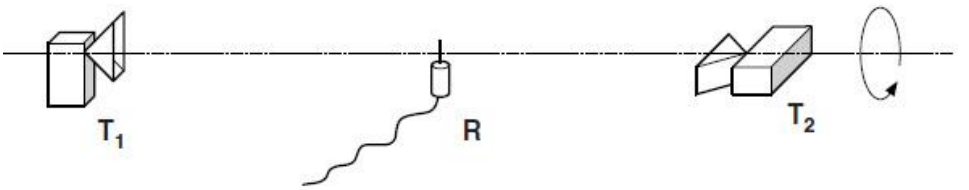

Fig. 3.2

The student observes that the intensity of the signal at R is halved.

The detected signal remains the same when R is moved 0.75 cm towards a transmitter.

Explain these observations.

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..... [3]